

# Chance Song

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## Education

**University of British Columbia** | *B.Sc Computer Science* | 3.9/4 GPA

Vancouver, BC 2024-2028

- Relevant Coursework: Data Structures and Algorithms, Software Engineering, Database Systems, Computer Systems, Data Science, Statistics, Linear Algebra, Discrete Mathematics, Models of Computation, Object-Oriented Design

## Projects

**AllBookd** | *TypeScript, Next.js, React, Supabase, PostgreSQL, Tailwind CSS, Framer Motion, Vercel*

2025 – 2026

- Shipped a full-stack web app for discovery and tailored reading using Next.js, React and Supabase, deployed live on Vercel.
- Built authentication and cross-device session sync using Supabase Auth and PostgreSQL row-level security on user tables.
- Designed responsive UI in Tailwind CSS and Framer Motion while improving usability through iterative Figma prototyping.
- Cut initial page load time by 45% by migrating client-rendered routes to server-side rendering with response-level caching.
- Modeled catalog and user-library data in a normalized PostgreSQL schema with indexed queries backing search and browse.

**LearnIO** | *HackCamp | Python, TypeScript, FastAPI, Electron, SQLite, Tailwind CSS, Gemini API, REST APIs*

2025

- Engineered a working end-to-end product within HackCamp's 24-hour build window by architecting an Electron and FastAPI system integrating Gemini inference, PDF processing, web scraping, and a local SQLite database.
- Cut time searching for practice exam from possibly hours to 30 seconds, by downloading PDFs and running Gemini extraction concurrently with asyncio, with a semaphore and exponential-backoff retries to stay within free-tier rate limits.
- Parsed, classified, and indexed 1,000+ exam questions with automated Python scraping pipelines built on FastAPI and SQLite.

**Probabilistic Poker Analytics Engine** | *Java, JUnit, Java Swing, JSON, OOP, UML*

2025

- Delivered win probabilities for up to 9 players in under 200 ms by building a Monte Carlo engine running 30,000+ trials.
- Cut per-evaluation runtime by 75% by replacing pairwise hand comparison with a precomputed hand-rank lookup table.
- Verified hand-ranking logic across all 2,598,960 possible five-card hands with 70+ JUnit tests reaching 100% line coverage.
- Simplified complex multi-player setup by drafting the system architecture in UML and building an interactive Java Swing UI.

**Mobile Android Game App** | *Godot, GDScript, Agile development, Kanban*

2024 – 2025

- Shipped a downloadable Android game with physics-driven movement, collision handling, and level progression in Godot.
- Structured game logic around a reusable object-oriented entity system with persistent save state and data-driven levels.
- Designed an idle-economy loop with exponential cost scaling across 4 upgrade paths, driving long-term player progression.

## Experience and Leadership

**Ante** | *Software Engineer*

Vancouver, BC 2026

- Engineered an android bankroll tracking app in React Native, establishing a streamlined session ledger for users to track with per-session ROI, win rate, and long-run performance analytics that replace manual spreadsheet tracking.
- Surfaced behavioral patterns invisible to recall like conditional win rates gated by sample size, loss-chasing detection from post-loss bet sizing, and a consistency score over full session history.
- Eliminated a silent bug that dropped synced sessions between accounts, by building offline-first Supabase sync with encrypted local storage and integration tests that reproduced the clock-skew failure.
- Caught state-sync race conditions in multi-session and multi-account flows before release through 240 Jest tests across 14 suites exercising real context and hook logic.

**UBC Geering Up** | *Junior Instructor*

Vancouver, BC 2024

- Taught programming, object-oriented design, and game development to 60+ students across 12 week-long summer camps.
- Raised project completion rates by mentoring students through live debugging and structured problem-solving sessions.

**Programming Club** | *Club President*

Port Moody, BC 2021 – 2024

- Founded and led a programming club with 50+ active members, overseeing recruitment, workshops, and weekly lessons.
- Organized and hosted competitive programming contests including CCC and BCC, drawing hundreds of student participants.

## Technical Skills

**Languages:** Python, Java, TypeScript/JavaScript, SQL, C, C++, Go, Kotlin

**Technologies:** SQLite, PostgreSQL, FastAPI, Docker, React, Tailwind CSS, PyTorch, Vercel, Figma, Scrapy, Git, AWS, Expo, Jest

**Concepts:** System Design, CI/CD, DevOps, Backend, Full Stack, Agile, REST APIs, Unit testing, Version control, Automated Testing